

$\therefore (p \wedge q \wedge r) \vee (p \wedge \neg q \wedge r) \vee (\neg p \wedge q \wedge r)$ is True $\Rightarrow$ Only two of p, q, r is True.

Proof of converse

Case 1: p is True, q is True, r is False. By the defn. of negation, $\neg r$ is True.
By the defn. of AND, $(p \wedge q \wedge \neg r)$ is True.

By the defn. of OR, $(p \wedge q \wedge \neg r) \vee (p \wedge \neg q \wedge r) \vee (\neg p \wedge q \wedge r)$ is True.

Case 2: p is True, q is False, r is True. Similar proof to Case 1

Case 3: p is False, q is True, r is True. Similar proof to Case 1

Only two of p, q, r is True $\Rightarrow (p \wedge q \wedge \neg r) \vee (p \wedge \neg q \wedge r) \vee (\neg p \wedge q \wedge r)$ is True $\square$

46) Suppose that a truth table in n propositional variables is specified. Show that a compound proposition with this truth table can be formed by taking the disjunction of conjunctions of the variables or their negations, with one conjunction included for each combination of values for which the compound proposition is true. The resulting compound proposition is said to be in disjunctive normal form.

Soln: Suppose, we have a truth table having n propositional variables $p_1, p_2, \dots, p_n$. There are 2^n rows in the truth table (mentioned in the book). For each row in the table, where the final compound proposition column has a truth value True, Take the conjunctions of the propositional variables in the following manner: $(p_1 \wedge p_2 \wedge \dots \wedge p_i \wedge \dots \wedge p_n)$

For all such p_i , where p_i has truth value False for that row, take $\neg p_i$ in the conjunction since by the defn. of negation, $\neg p_i$ has truth value True. For all such p_i , where p_i has truth value True for that row, we take p_i in the conjunction (self-explanatory)

By defn. of AND, it's easy to see why such a conjunction is True for each row. For all such rows, ~~and~~ ^{take} the disjunction of each such conjunction where the compound proposition has truth value True for the given row.

By the defn. of OR, such a disjunction is True $\square$